

CPCS212: Numerical Methods

Week11

Numerical Integration



Lecture 24

Introduction to Numerical Integration



- Definitions
- Upper and Lower Sums
- Trapezoid Method (Newton-Cotes Methods)
- Romberg Method
- Gauss Quadrature
- Examples

Integration

Indefinite Integrals

$$\int x \, dx = \frac{x^2}{2} + c$$

Indefinite Integrals of a function are functions that differ from each other by a constant.

Definite Integrals

$$\int_0^1 x \, dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}$$

Definite Integrals are numbers.

Fundamental Theorem of Calculus

If f is continuous on an interval $[a, b]$,

F is antiderivative of f (i.e., $F'(x) = f(x)$)

$$\int_a^b f(x) dx = F(b) - F(a)$$

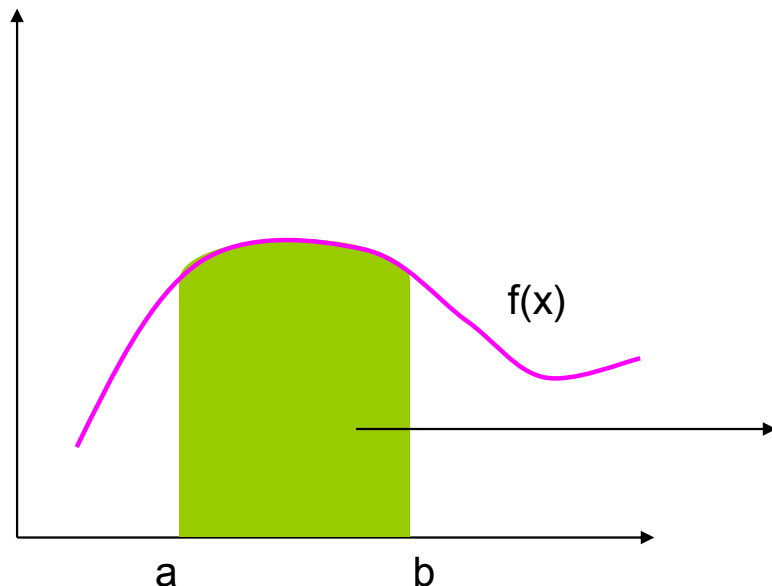
There is no antiderivative for : e^{x^2}

No closed form solution for : $\int_a^b e^{x^2} dx$

The Area Under the Curve

One interpretation of the definite integral is:

Integral = area under the curve



$$Area = \int_a^b f(x) dx$$

Upper and Lower Sums

The interval is divided into subintervals.

$$\text{Partition } P = \{a = x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n = b\}$$

Define

$$m_i = \min \{f(x) : x_i \leq x \leq x_{i+1}\}$$

$$M_i = \max \{f(x) : x_i \leq x \leq x_{i+1}\}$$

Lower sum

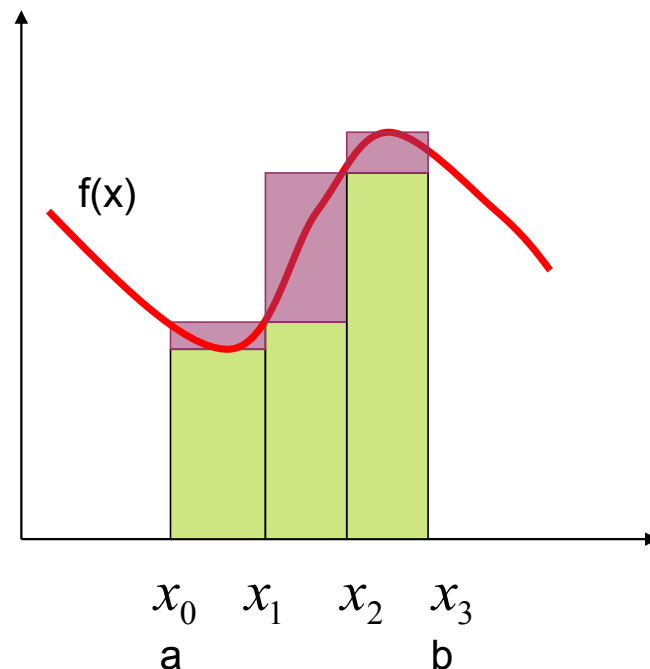


$$L(f, P) = \sum_{i=0}^{n-1} m_i (x_{i+1} - x_i)$$

Upper sum



$$U(f, P) = \sum_{i=0}^{n-1} M_i (x_{i+1} - x_i)$$



Upper and Lower Sums

Lower sum

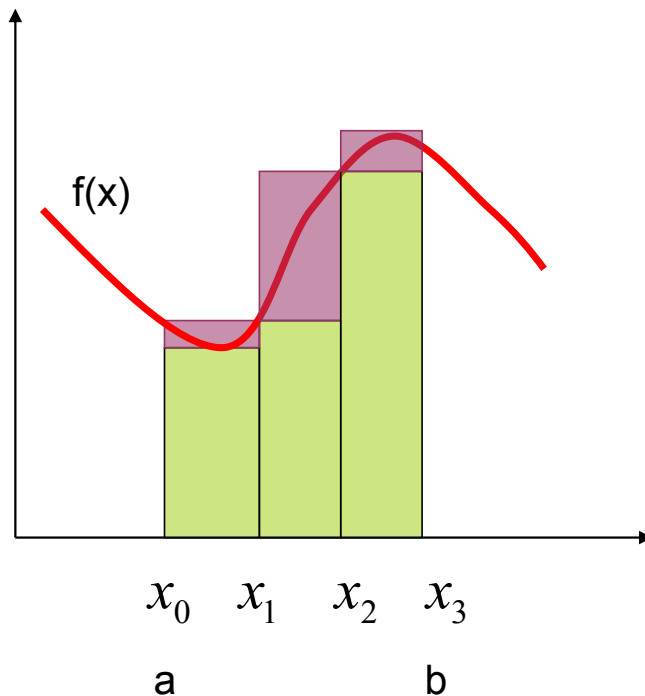
$$L(f, P) = \sum_{i=0}^{n-1} m_i (x_{i+1} - x_i)$$

Upper sum

$$U(f, P) = \sum_{i=0}^{n-1} M_i (x_{i+1} - x_i)$$

$$\text{Estimate of the integral} = \frac{L + U}{2}$$

$$\text{Error} \leq \frac{U - L}{2}$$



Example

$$\int_0^1 x^2 dx$$

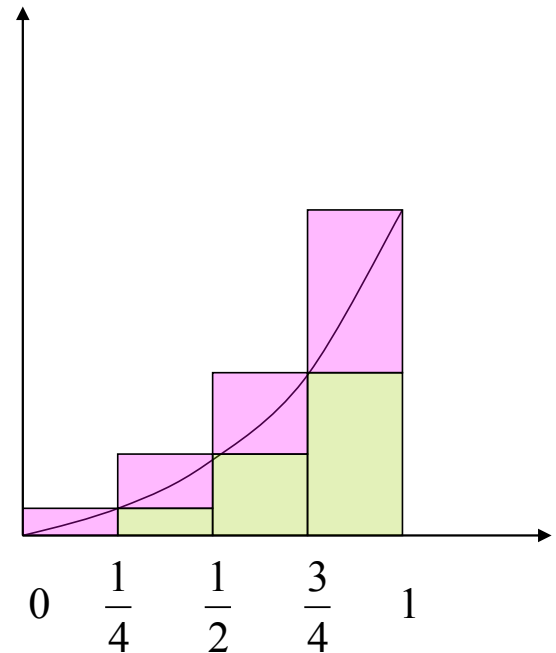
$$\text{Partition: } P = \left\{0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1\right\}$$

$n = 4$ (four equal intervals)

$$m_0 = 0, \quad m_1 = \frac{1}{16}, \quad m_2 = \frac{1}{4}, \quad m_3 = \frac{9}{16}$$

$$M_0 = \frac{1}{16}, \quad M_1 = \frac{1}{4}, \quad M_2 = \frac{9}{16}, \quad M_3 = 1$$

$$x_{i+1} - x_i = \frac{1}{4} \quad \text{for } i = 0, 1, 2, 3$$



Example

$$\text{Lower sum} \quad L(f, P) = \sum_{i=0}^{n-1} m_i (x_{i+1} - x_i)$$

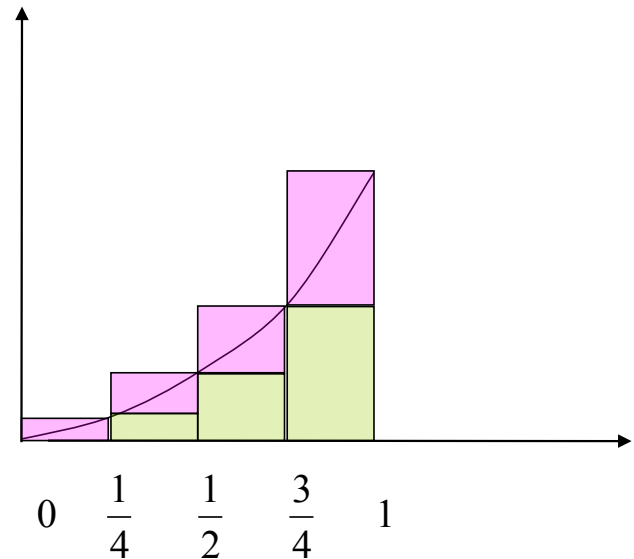
$$L(f, P) = \frac{1}{4} \left[0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16} \right] = \frac{14}{64}$$

$$\text{Upper sum} \quad U(f, P) = \sum_{i=0}^{n-1} M_i (x_{i+1} - x_i)$$

$$U(f, P) = \frac{1}{4} \left[\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1 \right] = \frac{30}{64}$$

$$\text{Estimate of the integral} = \frac{1}{2} \left(\frac{30}{64} + \frac{14}{64} \right) = \frac{11}{32}$$

$$\text{Error} < \frac{1}{2} \left(\frac{30}{64} - \frac{14}{64} \right) = \frac{1}{8}$$



Upper and Lower Sums

- Estimates based on Upper and Lower Sums are easy to obtain for **monotonic** functions (**always increasing or always decreasing**).
- For non-monotonic functions, finding maximum and minimum of the function can be difficult and other methods can be more attractive.

Newton-Cotes Methods

- In **Newton-Cote Methods**, the function is approximated by a **polynomial of order n** .
- Computing the integral of a polynomial is easy.

$$\int_a^b f(x)dx \approx \int_a^b (a_0 + a_1x + \dots + a_nx^n)dx$$

$$\int_a^b f(x)dx \approx a_0(b-a) + a_1 \frac{(b^2 - a^2)}{2} + \dots + a_n \frac{(b^{n+1} - a^{n+1})}{n+1}$$

Newton-Cotes Methods

- Trapezoid Method (First Order Polynomials are used)

$$\int_a^b f(x)dx \approx \int_a^b (a_0 + a_1x)dx$$

- Simpson 1/3 Rule (Second Order Polynomials are used)

$$\int_a^b f(x)dx \approx \int_a^b (a_0 + a_1x + a_2x^2)dx$$

Trapezoid Method

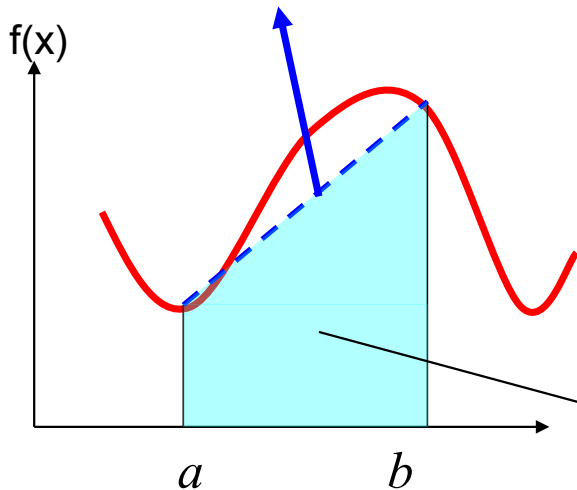


- ❑ Derivation-One Interval
- ❑ Multiple Application Rule
- ❑ Estimating the Error
- ❑ Recursive Trapezoid Method

Read 21.1

Trapezoid Method

$$f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$



$$I = \int_a^b f(x) dx$$

$$I \approx \int_a^b \left(f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right) dx$$

$$= \left(f(a) - a \frac{f(b) - f(a)}{b - a} \right) x \Big|_a^b$$

$$+ \frac{f(b) - f(a)}{b - a} \frac{x^2}{2} \Big|_a^b$$

$$= (b - a) \frac{f(b) + f(a)}{2}$$

Trapezoid Method

Derivation-One Interval

$$I = \int_a^b f(x)dx \approx \int_a^b \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right) dx$$

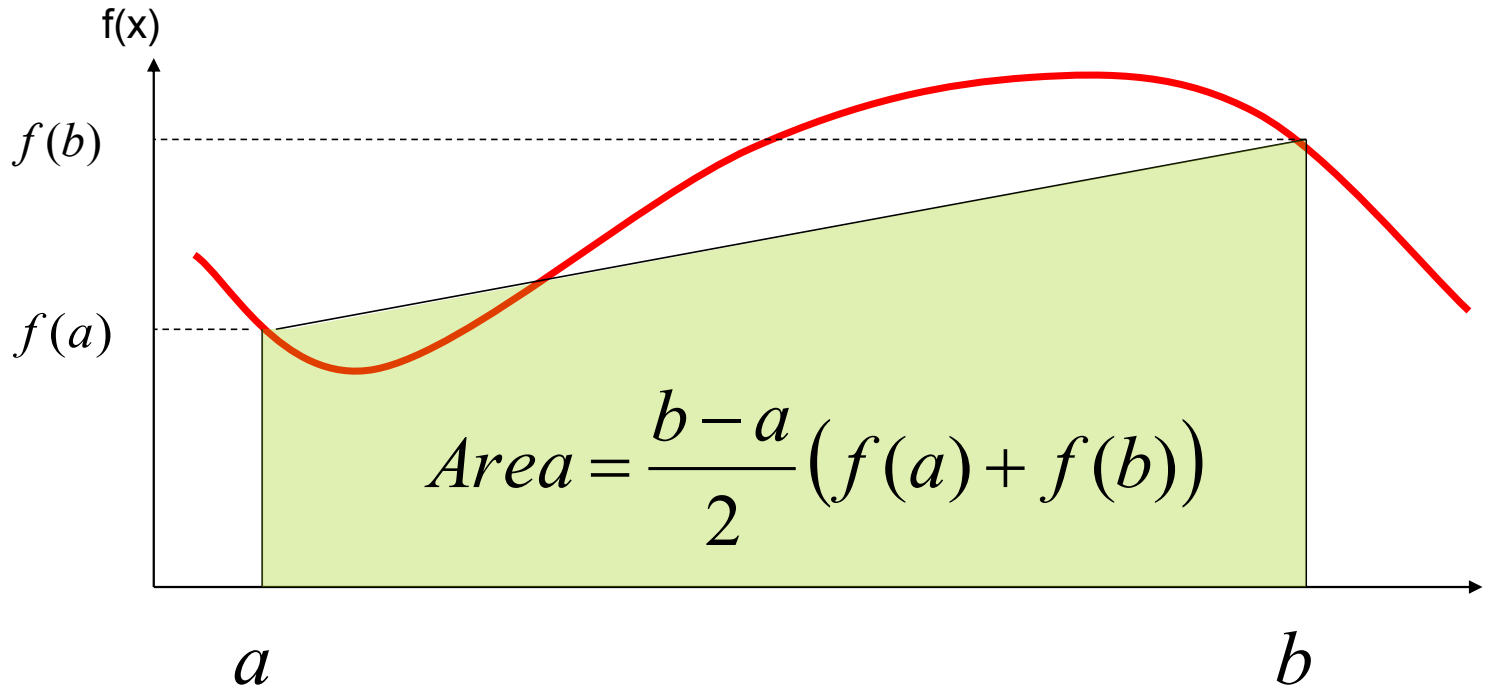
$$I \approx \int_a^b \left(f(a) - a \frac{f(b) - f(a)}{b - a} + \frac{f(b) - f(a)}{b - a} x \right) dx$$

$$= \left(f(a) - a \frac{f(b) - f(a)}{b - a} \right) x \Big|_a^b + \frac{f(b) - f(a)}{b - a} \frac{x^2}{2} \Big|_a^b$$

$$= \left(f(a) - a \frac{f(b) - f(a)}{b - a} \right) (b - a) + \frac{f(b) - f(a)}{2(b - a)} (b^2 - a^2)$$

$$= (b - a) \frac{f(b) + f(a)}{2}$$

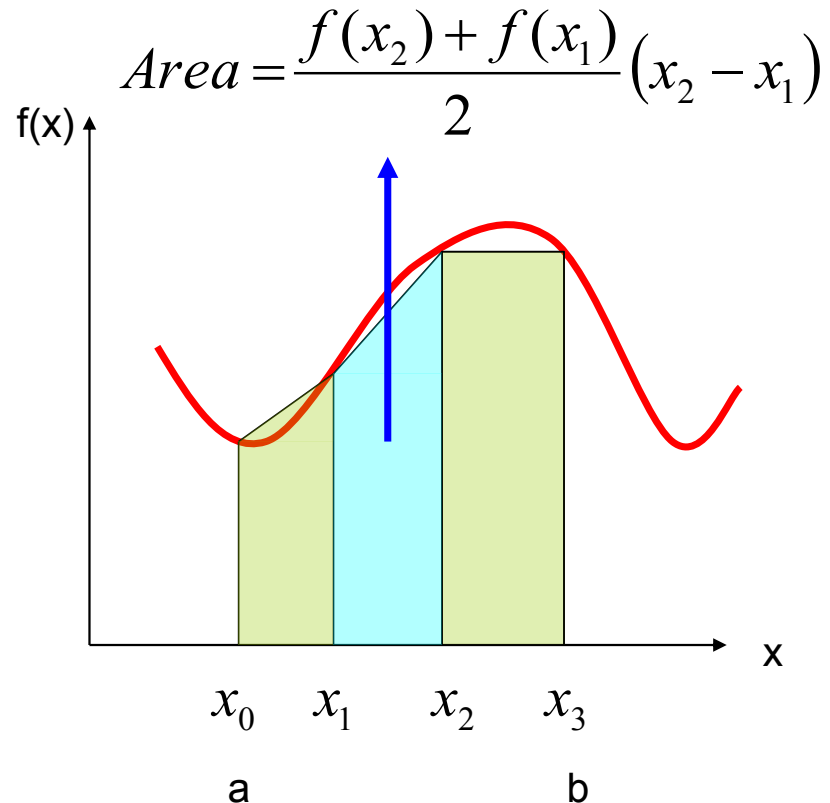
Trapezoid Method



Trapezoid Method

Multiple Application Rule

The interval $[a, b]$ is partitioned into n segments
 $a = x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n = b$
 $\int_a^b f(x) dx = \text{sum of the areas of the trapezoids}$



Trapezoid Method

General Formula and Special Case

If the interval is divided into n segments (not necessarily equal)

$$a = x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n = b$$

$$\int_a^b f(x)dx \approx \sum_{i=0}^{n-1} \frac{1}{2}(x_{i+1} - x_i)(f(x_{i+1}) + f(x_i))$$

Special Case (Equally spaced base points)

$$x_{i+1} - x_i = h \quad \text{for all } i$$

$$\int_a^b f(x)dx \approx h \left[\frac{1}{2}[f(x_0) + f(x_n)] + \sum_{i=1}^{n-1} f(x_i) \right]$$

Example

Given a tabulated values of the velocity of an object.

Time (s)	0.0	1.0	2.0	3.0
Velocity (m/s)	0.0	10	12	14

Obtain an estimate of the distance traveled in the interval $[0,3]$.

Distance = integral of the velocity

$$\text{Distance} = \int_0^3 V(t) dt$$

Example 1

The interval is divided
into 3 subintervals
Base points are $\{0,1,2,3\}$

Time (s)	0.0	1.0	2.0	3.0
Velocity (m/s)	0.0	10	12	14

Trapezoid Method

$$h = x_{i+1} - x_i = 1$$

$$T = h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} (f(x_0) + f(x_n)) \right]$$

$$\text{Distance} = 1 \left[(10 + 12) + \frac{1}{2} (0 + 14) \right] = 29$$

Estimating the Error

For Trapezoid Method

How many equally spaced intervals are needed to compute $\int_0^{\pi} \sin(x) dx$ to 5 decimal digit accuracy?

Error in estimating the integral

Theorem

Assumption : $f''(x)$ is continuous on $[a, b]$

Equal intervals (width = h)

Theorem: If Trapezoid Method is used to

approximate $\int_a^b f(x)dx$ then

$$Error = -\frac{b-a}{12} h^2 f''(\xi) \quad \text{where } \xi \in [a, b]$$

$$|Error| \leq \frac{b-a}{12} h^2 \max_{x \in [a, b]} |f''(x)|$$

Example

$$\int_0^{\pi} \sin(x) dx, \quad \text{find } h \text{ so that } |\text{error}| \leq \frac{1}{2} \times 10^{-5}$$

$$|\text{Error}| \leq \frac{b-a}{12} h^2 \max_{x \in [a,b]} |f''(x)|$$

$$b = \pi; \quad a = 0; \quad f'(x) = \cos(x); \quad f''(x) = -\sin(x)$$

$$|f''(x)| \leq 1 \quad \Rightarrow \quad |\text{Error}| \leq \frac{\pi}{12} h^2 \leq \frac{1}{2} \times 10^{-5}$$

$$\Rightarrow \quad h^2 \leq \frac{6}{\pi} \times 10^{-5}$$

Example

x	1.0	1.5	2.0	2.5	3.0
f(x)	2.1	3.2	3.4	2.8	2.7

Use Trapezoid method to compute: $\int_1^3 f(x)dx$

$$\text{Trapezoid } T(f, P) = \sum_{i=0}^{n-1} \frac{1}{2} (x_{i+1} - x_i) (f(x_{i+1}) + f(x_i))$$

Special Case: $h = x_{i+1} - x_i$ for all i ,

$$T(f, P) = h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} (f(x_0) + f(x_n)) \right]$$

Example

x	1.0	1.5	2.0	2.5	3.0
f(x)	2.1	3.2	3.4	2.8	2.7

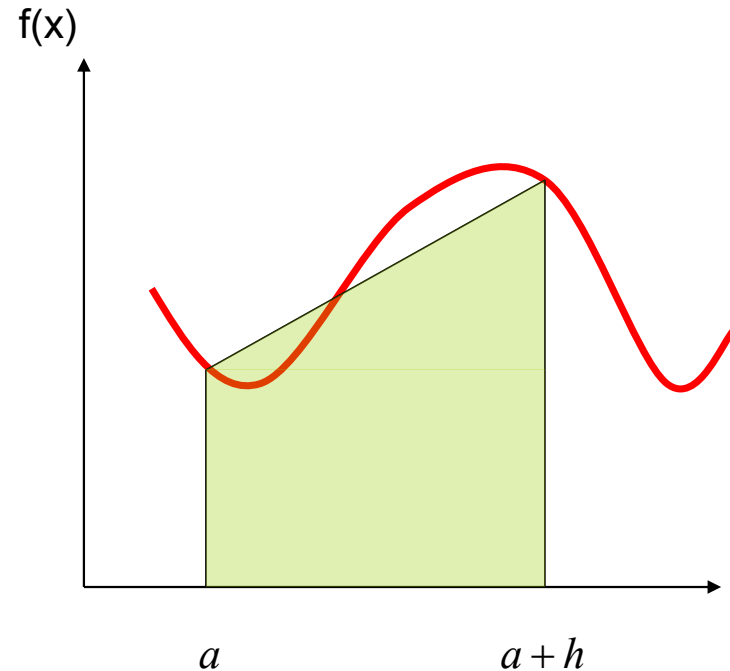
$$\begin{aligned}\int_1^3 f(x)dx &\approx h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2}(f(x_0) + f(x_n)) \right] \\ &= 0.5 \left[3.2 + 3.4 + 2.8 + \frac{1}{2}(2.1 + 2.7) \right] \\ &= 5.9\end{aligned}$$

Recursive Trapezoid Method

Estimate based on one interval:

$$h = b - a$$

$$R(0,0) = \frac{b-a}{2} (f(a) + f(b))$$



Recursive Trapezoid Method

Estimate based on 2 intervals:

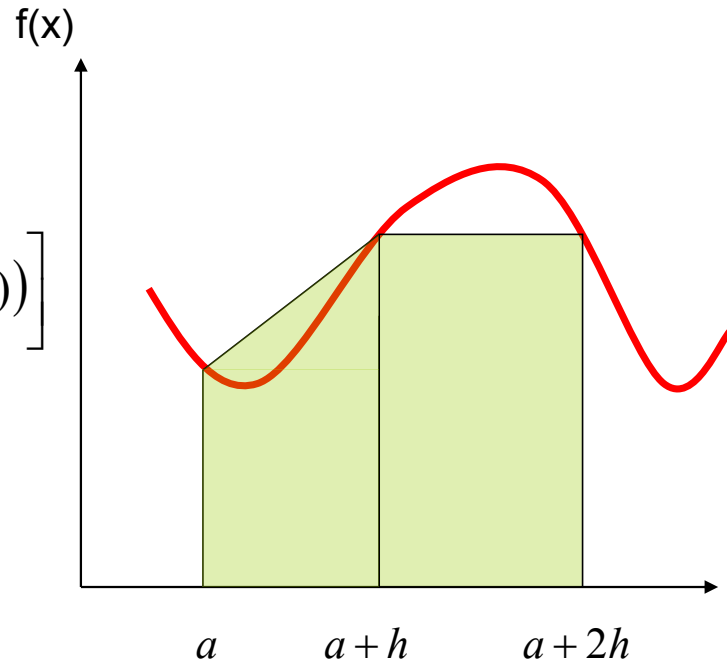
$$h = \frac{b-a}{2}$$

$$R(1,0) = \frac{b-a}{2} \left[f(a+h) + \frac{1}{2}(f(a) + f(b)) \right]$$

$$R(1,0) = \frac{1}{2} R(0,0) + h[f(a+h)]$$

Based on previous estimate

Based on new point



Recursive Trapezoid Method

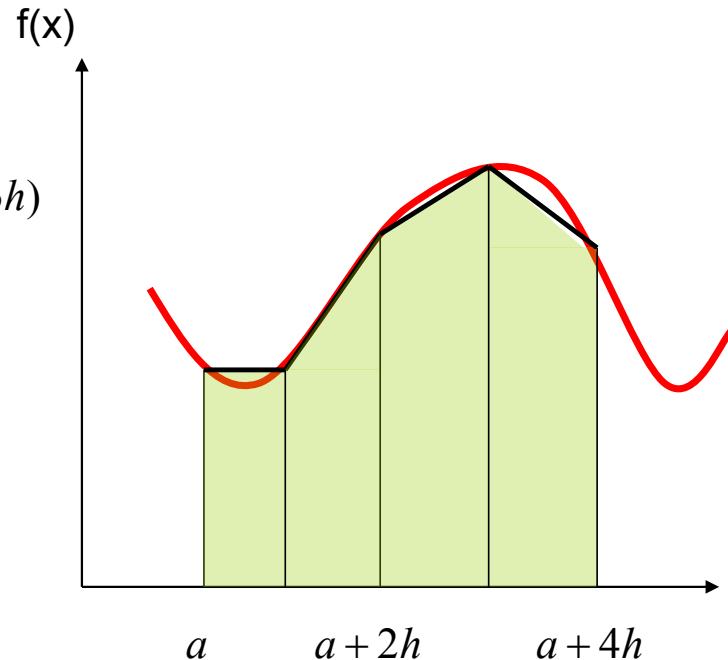
$$h = \frac{b-a}{4}$$

$$R(2,0) = \frac{b-a}{4} \left[f(a+h) + f(a+2h) + f(a+3h) + \frac{1}{2}(f(a) + f(b)) \right]$$

$$R(2,0) = \frac{1}{2} R(1,0) + h[f(a+h) + f(a+3h)]$$

Based on previous estimate

Based on new points



Recursive Trapezoid Method

Formulas

$$R(0,0) = \frac{b-a}{2} [f(a) + f(b)]$$

$$R(n,0) = \frac{1}{2} R(n-1,0) + h \left[\sum_{k=1}^{2^{(n-1)}} f(a + (2k-1)h) \right]$$

$$h = \frac{b-a}{2^n}$$

Recursive Trapezoid Method

$$h = b - a, \quad R(0,0) = \frac{b-a}{2} [f(a) + f(b)]$$

$$h = \frac{b-a}{2}, \quad R(1,0) = \frac{1}{2} R(0,0) + h \left[\sum_{k=1}^1 f(a + (2k-1)h) \right]$$

$$h = \frac{b-a}{2^2}, \quad R(2,0) = \frac{1}{2} R(1,0) + h \left[\sum_{k=1}^2 f(a + (2k-1)h) \right]$$

$$h = \frac{b-a}{2^3}, \quad R(3,0) = \frac{1}{2} R(2,0) + h \left[\sum_{k=1}^{2^2} f(a + (2k-1)h) \right]$$

.....

$$h = \frac{b-a}{2^n}, \quad R(n,0) = \frac{1}{2} R(n-1,0) + h \left[\sum_{k=1}^{2^{(n-1)}} f(a + (2k-1)h) \right]$$

Advantages of Recursive Trapezoid

Recursive Trapezoid:

- Gives the same answer as the standard Trapezoid method.
- Makes use of the available information to reduce the computation time.
- Useful if the number of iterations is not known in advance.

Motivation

Trapezoid Method:

$$\int_a^b f(x)dx = h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} (f(x_0) + f(x_n)) \right]$$

It can be expressed as:

$$\int_a^b f(x)dx = \sum_{i=0}^n c_i f(x_i)$$

$$\text{where } c_i = \begin{cases} h & i = 1, 2, \dots, n-1 \\ 0.5h & i = 0 \text{ and } n \end{cases}$$